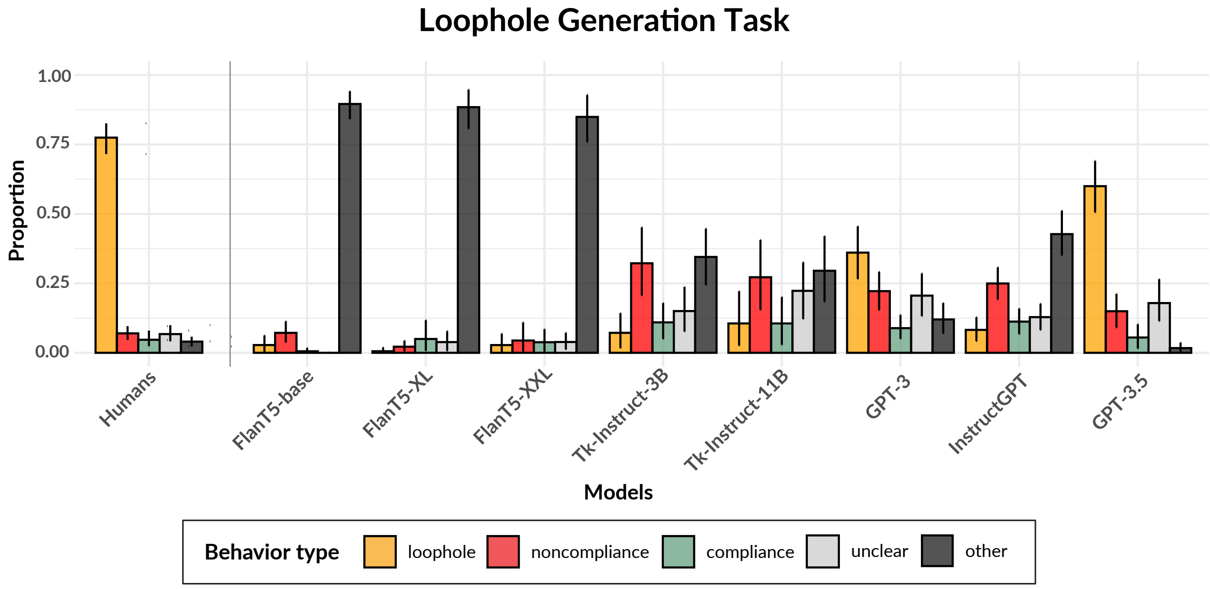

Figure 3: Comparison across humans and models on the distribution of the category of generated responses in the loophole generation task.

ratings to compliance, it also underestimates the trouble associated with both non-compliance and loopholes, relative to humans.

How upset is the other person about the protagonist's behavior? Similar to the trouble metric, humans rate compliance as resulting in almost no upset to the other person, and loophole behavior as resulting in less upset than non-compliance. GPT-3.5 once again comes closest to this baseline with loopholes resulting in less upset than non-compliance ($p = 0.014$). GPT-3 ($p = 0.002$) and Tk-Instruct-3B ($p = 0.017$) also perform reasonably well on this metric, with loopholes resulting in significantly less upset than non-compliance, but once again, overestimate the costs associated with compliance compared to the human baseline. InstructGPT and the smallest FlanT5 model all entirely fail to differentiate the behavior types on this metric, while Tk-Instruct-11B performs similarly, differentiating loopholes from non-compliance ($p = 0.019$), but not compliance from loopholes ($p = 0.676$). The remaining two models, FlanT5-XL and XXL almost get the relative amount of upset correct by rating compliance lower than loophole behavior, but then fail to significantly differentiate loopholes from non-compliance ($p = 0.186$, FlanT5-XL; $p = 0.132$, FlanT5-XXL).

How funny is the protagonist's behavior to the other person? Far fewer models effectively differentiate the behavior types for the metric of humor. While humans generally produce low ratings for all behavior types on this metric, they do reliably recognize some humor in the creative exploitation of loopholes. FlanT5-XXL, GPT-3, and GPT-3.5 all appear to have higher ratings for humor for loopholes than non-compliance, however this difference is not significant for any of the models ($p = 0.198$, FlanT5-XXL; $p = 0.240$, GPT-3; $p = 0.545$, GPT-3.5). Meanwhile, the 3 billion parameter Tk-Instruct model rates all of the behaviors as not funny, which could either reflect a lack of understanding about expectations or could be interpreted as a reasonable real-world response to an adult exploiting a loophole. The remaining models all fail to differentiate the three behavior types on this metric, with InstructGPT seeming to perform worst of all and generating almost entirely empty responses for all the behavior types (see Figure 4 in Appendix for information about the proportion of empty generations).

3.2 Loophole Generation

When it comes to loophole production, GPT-3.5 (loopholes = 59.9%) is the only model that approaches the human baseline (loopholes = 77.4%), with significantly more loopholes than all other response categories. GPT-3 comes close, with the majority of its generations being categorized as loophole, but still a high proportion of non-compliant and unclear actions. The two Tk-Instruct models fare slightly better than the best InstructGPT model, in that they produce the fewest number of incoherent/irrelevant generations. Still, the most of their remaining generations constitute non-compliance or involve lying or negotiation, but do not achieve the criteria of a loophole. All three of the FlanT5 models overwhelmingly generate incoherent or irrelevant responses ("other"), even though the two largest—XL and XXL—fared reasonably at aspects of the evaluation task, suggesting a separation between these abilities in these models. Table 2-5 in Appendix D showcase examples of generated loopholes from models and humans.

To better understand the shortcomings of the models on this task, we more closely examined the responses coded as "unclear." Unlike the "other" category, these responses were relevant and coherent, but often confused compliance and non-compliance without actually achieving the criteria of a loophole. Among the models with a significant proportion of this response type, GPT-3, GPT-3.5, and Tk-Instruct-3B had higher percentages of unclear responses that described negotiation between the protagonist and the other person in attempt to delay or avoid compliance (overall rate 13% for GPT-3, 11% for GPT-3.5, 7% for Tk-Instruct-3B). The other two models, InstructGPT and Tk-Instruct-11B, had higher percentages of unclear responses that described lying or deceitful actions, such as hiding and pretending (overall rate 12% for Tk-Instruct-11B, 11% for GPT-3.5, 6% for InstructGPT).

4 Discussion

The metrics that we use to assess models' ability to evaluate different behavior types—trouble, upset, and funny—can be thought of as representing two different aspects of costs and rewards. A more egocentric perspective is assessed by the trouble metric, where the models have to reason mostly about the direct consequences of an action. However, reasoning about the amount of upset those actions might cause to another person or how funny they might find such actions, requires higher-order, Theory of Mind reasoning. Further, among the latter two metrics, humor is thought to require especially complex reasoning about social conventions and expectations, with some theories suggesting that humor arises from a violation of our mental patterns and expectations (Deckers and Kizer, 1975). We find that the models' performance track on the evaluation task with the graded social and pragmatic complexity of these metrics. While most models are able to reason about the direct consequences of a protagonist's actions through the amount of trouble they would get into, fewer are able to perform the higher-order reasoning needed to differentiate these behaviors on upset, while none managed to do so for humor. These findings corroborate with those of recent work that have studied humor in LLMs (Hu et al., 2022; Jentzsch and Kersting, 2023), suggesting that models continue to struggle to reason about complex social conventions and expectations. While the phenomena we study have been found to be sensitive to a variety of cultural and demographic factors (Martin and Ford, 2018; Hashimoto et al., 2012) and would benefit from further exploration with more culturally diverse populations, the models we test are known to have been trained on text data that reflects a similar distribution of cultural backgrounds as the population that participated in our behavioral studies. Thus, the models' failures may reflect a lack of a deeper cognitive understanding of the expectations, social norms, and incongruities needed to reason about the social consequences of loopholes and echo the growing skepticism about recent models' Theory of Mind abilities (e.g. Shapira et al., 2023; Sap et al., 2022b; Ullman, 2023).

We also note some effects of model size and training. At 250M parameters, FlanT5-base is significantly smaller than any other model we tested, and also performs the worst across both the evaluation and generation tasks. In addition to their smaller parameter size compared to the GPT models, all the T5-based models we tested also have smaller data input than the GPT models, with T5's C4 corpus comprising just 60% of GPT-3's training data. When comparing the performance of our two families of T5-based models, Tk-Instruct's fine-tuning on the larger and more diverse SUP-NATINST benchmark (compared to the FlanT5 task set) appears to translate to improvements on our task. Still, there appear to be interesting exceptions to the benefits of scaling on our task. For example, InstructGPT shows comparable performance to the two Tk-Instruct models on the generation task, despite being significantly larger than either (175B parameters vs. 3B and 11B). This could be because InstructGPT was specifically fine-tuned to improve alignment and better follow instructions (`https://openai.`

com/research/instruction-following), suggesting that instruction fine-tuning may not necessarily improve the flexibility of pragmatic reasoning.

Our study adds to a growing body of work evaluating LLMs' understanding of various pragmatic phenomena (Le et al., 2019; Hu et al., 2022; Valmeekam et al., 2022; Sap et al., 2022a; Ruis et al., 2022; Fried et al., 2022). Some of these phenomena, like deceit (Hu et al., 2022), approach the spirit of loopholes by probing understanding of misaligned values. Similar to the evaluation of conversational implicature in Ruis et al. (2022), our evaluation task also probes understanding of intentions, while additionally testing the costs and values associated with agreeing or refusing to comply with them, analogous to recent social commonsense reasoning benchmarks (Sap et al., 2022a). Our generation task goes beyond any of these formats to probe models' ability to produce pragmatic behavior, as opposed to choosing between answer options. When given information detailing the misalignment between the goals of different agents (e.g. the other person wants X, the protagonist wants Y), we assess whether models are able to generate reasonable actions that fall in the grey area between full compliance and outright non-compliance.

5 Conclusion

In this work, we compared the performance of several different LLMs to humans on two tasks designed to assess loophole comprehension. In the **evaluation** task, we find that a number of models, of varying sizes, are able to differentiate compliant, non-compliant, and loophole behaviors based on how much trouble the protagonist would get in for their action. However, fewer models are able to effectively differentiate these behavior types when it comes to reasoning about *others'* emotional reactions in response to such behaviors. We find that only three of the models, GPT-3.5, GPT-3, and Tk-Instruct-3B reflect human baselines on the metric of how upset the other person would be about the protagonist's behavior, in that they significantly differentiated loopholes from non-compliance and rated compliance with low levels of upset, with GPT-3.5 performing best . For the final metric—how funny the protagonist's behavior would be to the other person—no model was able to recognize the humor in the creative exploitation of loopholes in the way that humans do. When it comes to loophole **generation**, we find that only GPT-3.5 approaches the human baseline, with loopholes far outnumbering all other response categories. GPT-3 comes close, with slightly more loopholes than non-compliant actions. In the future, we are interested in how loophole behavior can be used as a testbed for developing techniques that could equip smaller models with similar human-like pragmatic reasoning capacity.

Limitations

With the exception of the 175B parameter OpenAI models that were accessible via their API, our own limited computational resources only allowed us to test models up to 11B parameters in size. Significant performance improvement on all tasks studied in this work may be achieved with increasing model sizes in the unstudied range beyond 11B parameters. Additionally, while we believe a low temperature setting is mostly appropriate for the question-answer format of the behavior evaluation task, the lack of an exhaustive parameter search for the more open-ended generation task may underestimate model capacity. However, we think that using the default temperature parameter in the text generation process still enables a fair and generalizable comparison across models.

Ethics Statement

A primary ethical concern of the present work is its engagement with a complex reasoning ability in machines that might lead to problematic consequences: the ability to exploit loopholes. Our assessment of LLMs capacity for loophole generation gives us a baseline understanding of how these models might be exploited for malicious purposes so that users can be better prepared for their interactions with these systems, and so that model developers can preemptively build the necessary guardrails against such vulnerabilities.

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